

Okan Berhoğlu

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Profile

Robotics Software Engineer experienced in desktop and industrial software development, machine data acquisition, device communication, and engineering systems. Skilled in communication protocols, database integration, deployment, validation, and technical documentation. Background in Mechanical Engineering and ongoing M.S. studies in Robotics.

Experience

Middle East Technical University

Ankara, Turkey

Researcher

October 2025 - Present

- Conduct research on mobile robotic systems, including kinematic and dynamic modeling, simulation, sensor integration, and autonomous control algorithm development.
- Contribute to a research project investigating human rhythmic behavior. Within the project, I developed an experimental setup using a 3-axis haptic device, collected data from predefined rhythmic motions, and conducted system identification and behavior modeling studies.

Teknopar Industrial Automation

Ankara, Turkey

Software Engineer

February 2024 – October 2025, 1 year 8 months

- Designed the software architecture and developed a Windows service-based data acquisition application for CNC controllers. Integrated TCP/UDP-based industrial communication protocols to collect real-time machine data, standardized the collected data into an MTConnect-like format, and actively contributed to the software's licensing, deployment, and commissioning processes.
- Developed an OSGi-based application with a Swing user interface for military truck systems, enabling operators to monitor onboard power units and control the vehicle's electrical system. Integrated industrial communication protocols such as EtherNet/IP and Modbus TCP for device communication and data collection.
- Actively contributed to the proposal writing of a TÜBİTAK-funded project focused on developing the system model, autonomy stack, and digital twin of a mobile robot. Within the project, worked on AI-based environmental perception and autonomous navigation using ROS2 as a software framework, 3D LiDAR, and depth camera sensors.

Candidate Engineer

December 2022 – February 2024, 1 year 3 months

- Supported R&D activities focused on collecting data from CNC machines and industrial systems.
- Contributed to the modeling of an electromechanical system using Simulink and Simscape.
- Analyzed experimental data from hardware prototypes and supported grey-box system identification and parameter estimation studies.
- Contributed to the integration of the developed system model into IBM Rhapsody, gaining practical experience in model integration and systems engineering workflows.

Intern

May 2022 – December 2022, 8 months

- Worked on data acquisition methods from CNC machines.

Anova R&D Technologies

Ankara, Turkey

Intern

May 2021 – August 2021, 4 months

- Gained hands-on experience in machining, quality control, and finishing processes, and contributed to mold and fixture design activities.

Education

Middle East Technical University

Ankara, Turkey

Master of Science, Robotics

2024 – 2026

TOBB Economics and Technology University

Ankara, Turkey

Bachelor of Science, Mechanical Engineering

2018 – 2023

Atatürk University

Remote

Associate degree, Computer Programming, Second University Program

2021 – 2023

Skills

Software: Java, C#, Python, C++, REST API, Git, Linux, CUDA, Unity, ROS/ROS2, MATLAB, Simulink, Simscape, SQL, InfluxDB

Industrial Communication: TCP/IP, UDP, EtherNet/IP, Modbus TCP, CAN, RS-485, industrial machine communication

Language: English - Upper-Intermediate, TOEFL iBT 2026 score: 5/6 (equivalent to 100/120 in old score system)